

Sales Forecasting & Demand Intelligence System

Executive Business Report

Prepared for: Head of Supply Chain & CFO — Period: 2015–2018 — Forecast Horizon: 3 Months

Executive Summary

This report summarizes a four-year analysis of superstore sales data covering over 9,800 transactions across three product categories and four regions. We built a forecasting system that predicts future sales with an error of just 4.7%, well within the industry standard of 15 to 25%. The system also detects unusual sales weeks automatically and groups products into four demand segments so the business can plan stock levels more intelligently. Sales are growing consistently, the West region and Office Supplies are the most reliable performers, and the biggest stocking risk right now is Machines which has been declining for three years.

Key Findings

Overall Business Growth

Total sales have grown consistently from 2015 to 2018, nearly doubling revenue in four years. The second half of each year drives the most revenue with November being the strongest month every year.

Category Performance

- Technology leads with the highest revenue at 827,000
- Furniture comes second at 728,000
- Office Supplies follows at 705,000

Regional Performance

West region nearly doubled from 146,000 in 2015 to 248,000 in 2018. East also grew well. Central stayed flat and South dipped in 2016 before recovering.

3-Month Sales Forecast

Model: Stacking Ensemble (MAPE 4.7%)

We tested four approaches. The Stacking model that combines SARIMA, Prophet and XGBoost performed best. For every 100 dollars predicted, the actual result is within 4 dollars and 70 cents.

Month	Forecast	Confidence
January 2019	\$78,718	High
February 2019	\$105,241	Medium
March 2019	\$84,427	Medium

February is forecast to be the strongest month, likely driven by post-January restocking and early quarter purchasing.

Unusual Sales Weeks Detected

Our system automatically flags weeks where sales behaved very differently from the expected pattern. Ten such weeks were detected across the four years.

Top 3 Anomalies and Their Likely Causes

1. February 2015 - Spike to 37,000 in one week

The single largest weekly spike in the entire dataset. A large bulk corporate order or an early promotional campaign most likely caused this. The sharp drop in the following two weeks confirms it was a one-time event, not a genuine demand shift.

2. August 2015 - Spike to 30,000

August typically sees elevated sales due to back to school purchasing and office restocking ahead of the new fiscal quarter. This is expected to repeat every year.

3. November 2017 and 2018 - Multiple spikes reaching 35,000

The strongest cluster of anomalies in the dataset. These align with Black Friday, end of year budget spending and corporate bulk buying before the year closes.

Product Demand Segments

Products were grouped into four demand clusters using KMeans clustering.

High Volume Stable - Chairs, Phones, Tables, Binders, Storage, Accessories. Core revenue drivers. Keep steady buffer stock.

High Growth Niche - Copiers only. Growing at 479% over four years. Stock carefully and monitor weekly.

Low Volume Steady - Art, Paper, Labels, Envelopes and others. Keep lean inventory with small regular orders.

Declining High Risk - Machines only. Down 29% since 2015. Reduce stock and evaluate discontinuation.

Business Recommendations

1. Invest in West Region and Office Supplies

Both showed the highest forecast accuracy and consistent growth. New marketing spend or capacity expansion should focus here first.

2. Pre-Stock Before November Every Year

November and December consistently produce the largest sales spikes. Pre-stocking High Volume Stable products by mid-October every year will prevent stockouts during peak weeks.

3. Move Budget from Machines to Copiers

Machines have declined three years straight while Copiers grew 479%. Reallocating shelf space and budget is the most data-backed decision available right now.

One Risk the Business Should Know

This system is built on four years of historical data. It cannot predict events that have never happened before - a new competitor, a supply chain disruption or a sudden economic shift. The forecast is a planning guide, not a guarantee. The team should review model accuracy every quarter and retrain it with fresh data. Technology sales showed a 46% forecast error which means that segment needs human judgment alongside the model output.

Prepared as part of a data science internship project. Analysis based on the Superstore Sales Dataset (2015–2018).